

# APACHE AIRFLOW: THE COMPLETE BEGINNER GUIDE



Apache  
**Airflow**

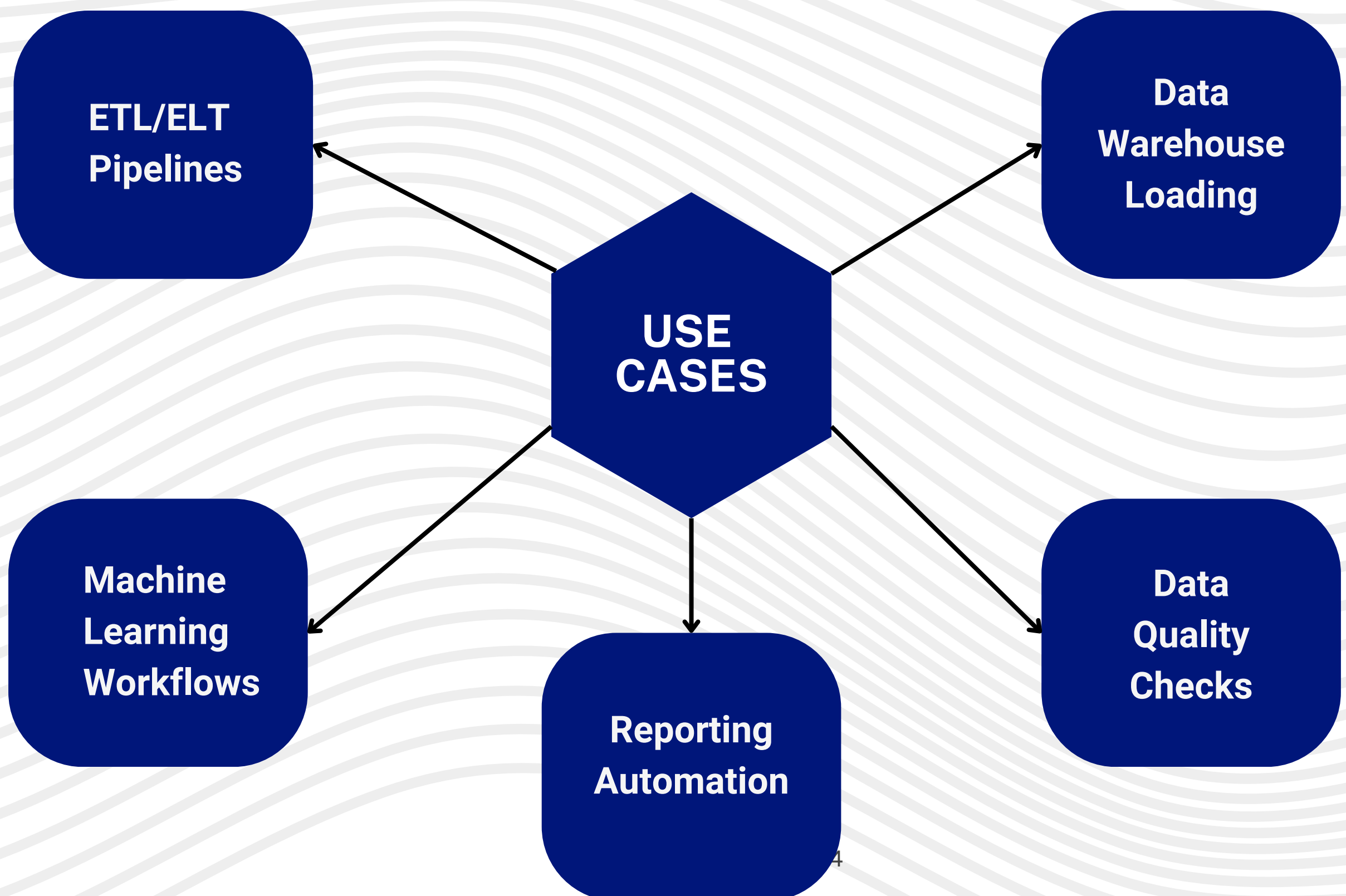
**BY MICHAEL FERNANDES**

# WHAT IS APACHE AIRFLOW

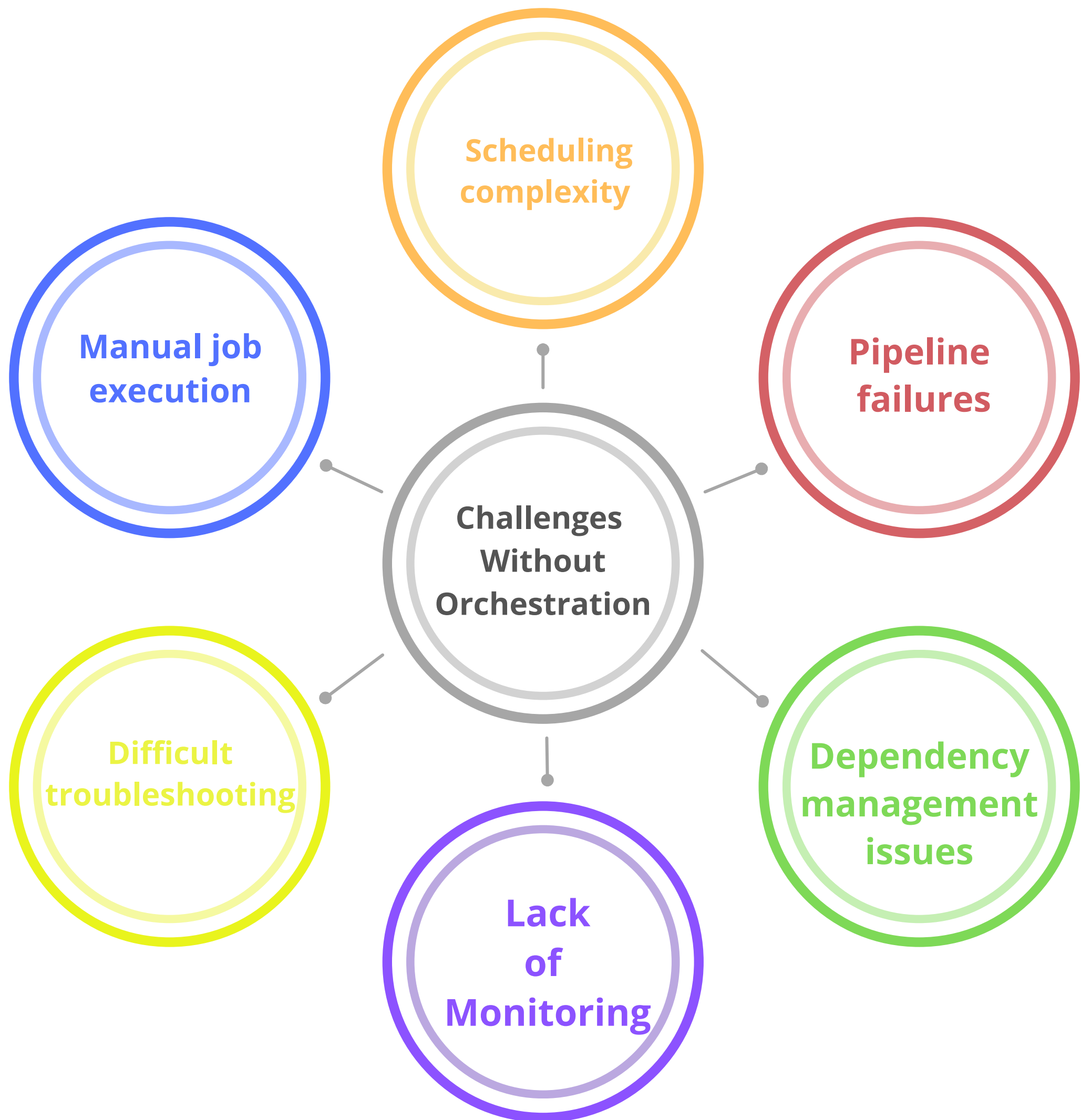
Apache Airflow is an open-source workflow orchestration platform used to automate, schedule, and monitor data pipelines.

Instead of manually running scripts and jobs, Airflow automates workflows and ensures tasks execute in the correct order.

Think of Airflow as: The Operating System for Data Pipelines



# WHY DO WE NEED AIRFLOW?



# HOW AIRFLOW SOLVES THESE PROBLEMS

## Integration with Modern Data Tools

Airflow connects seamlessly with technologies like Apache Spark, Kafka, Snowflake, dbt, BigQuery, Redshift, Databricks, and cloud services.

## Scalability

Airflow can orchestrate anything from a few daily jobs to thousands of workflows running across distributed infrastructure.

## Failure Detection & Alerts

Airflow automatically detects failed tasks and can send notifications via email, Slack, or other alerting systems for quick resolution.

## Workflow Automation

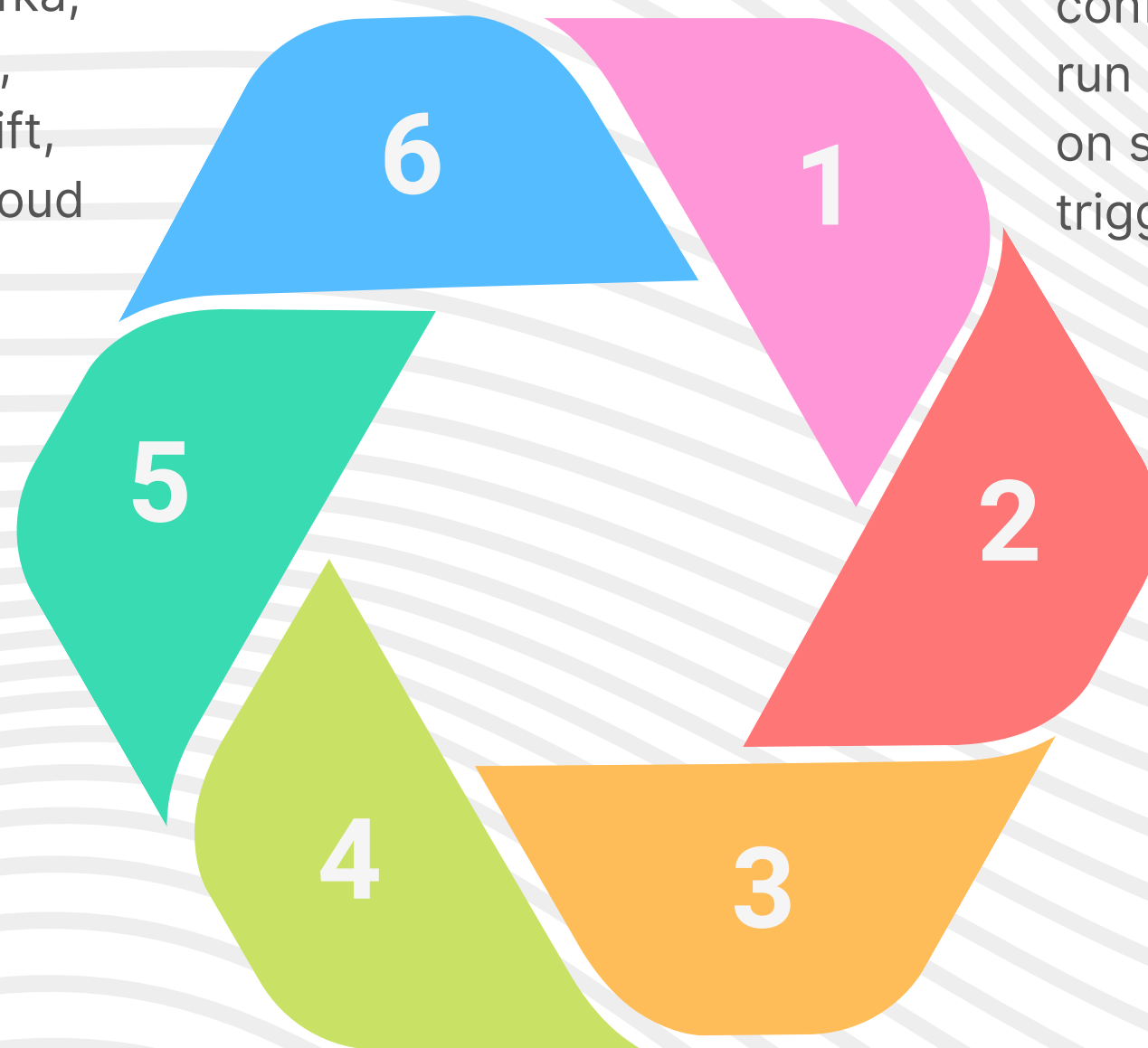
Airflow automates repetitive data tasks, eliminating the need for manual execution. Once configured, workflows run automatically based on schedules or triggers.

## Dependency Management

Airflow ensures tasks execute in the correct sequence. For example, data transformation won't start until data extraction has completed successfully.

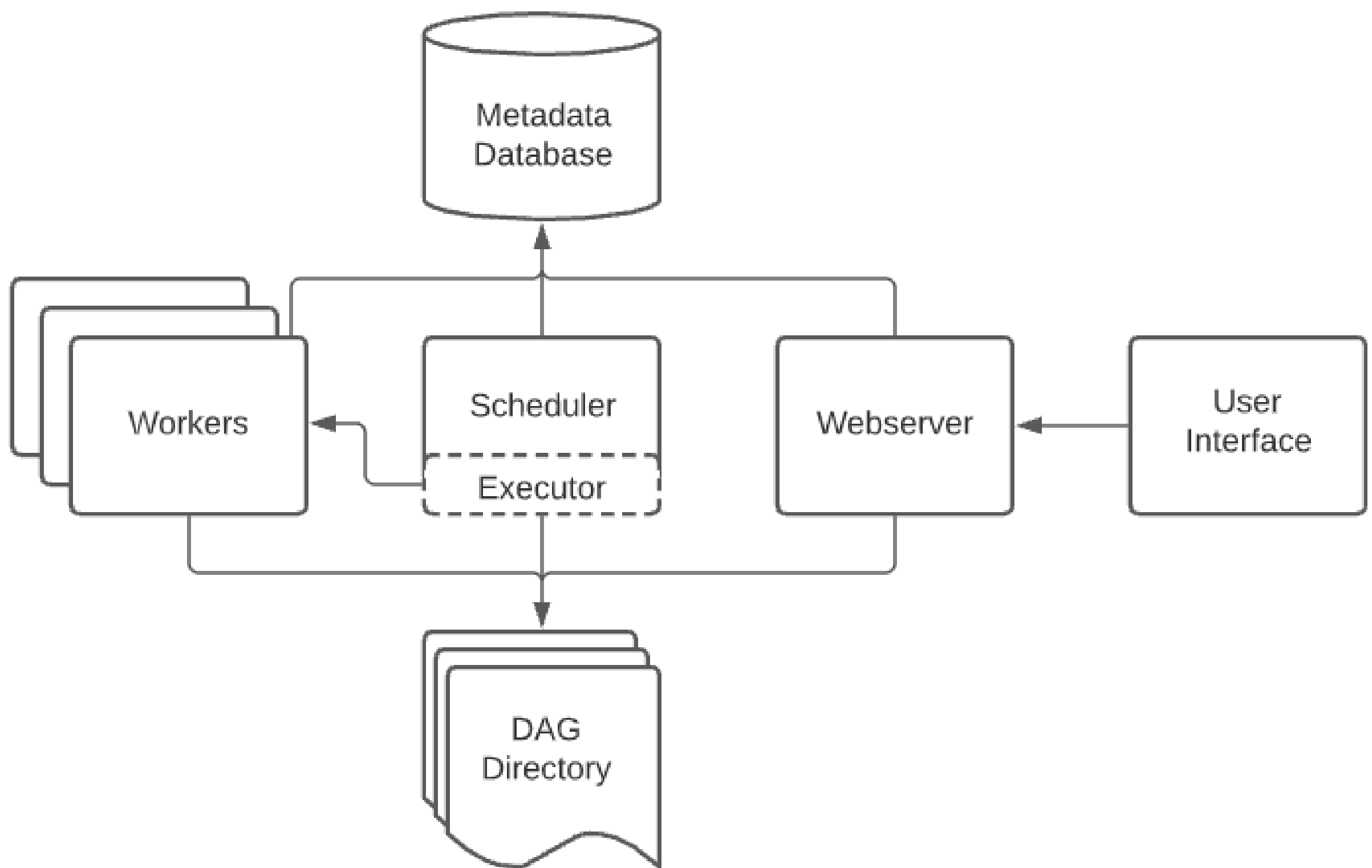
## Centralized Monitoring

The Airflow UI provides a single dashboard to track workflow status, task execution, logs, and historical runs in real time.





# AIRFLOW ARCHITECTURE



- DAG Directory → Stores workflow definitions (DAGs).
- Scheduler → Reads DAGs and decides which tasks should run.
- Executor → Assigns tasks to workers.
- Workers → Execute the actual tasks.
- Metadata Database → Stores DAGs, task status, logs, and execution history.
- Webserver → Provides the Airflow UI.
- User Interface → Lets users monitor, trigger, and manage workflows.

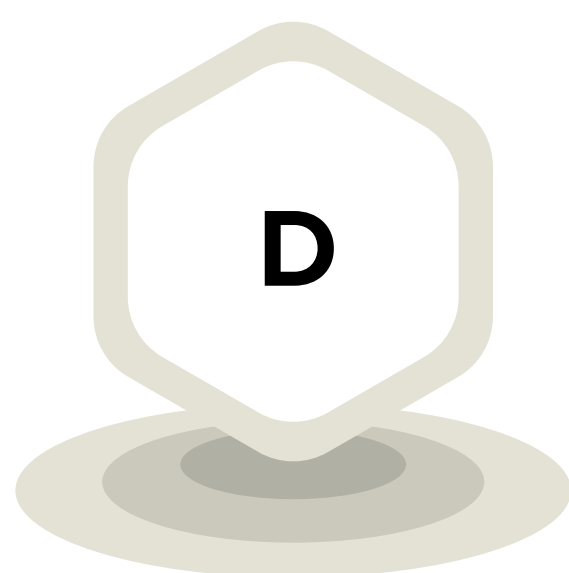
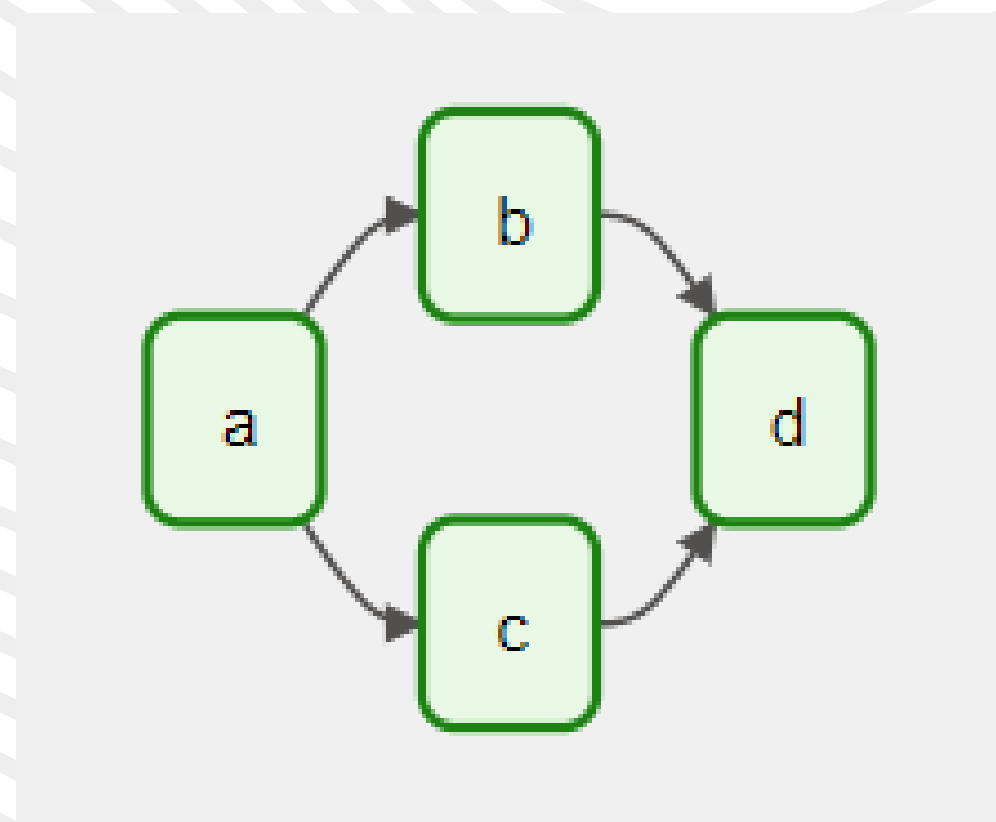
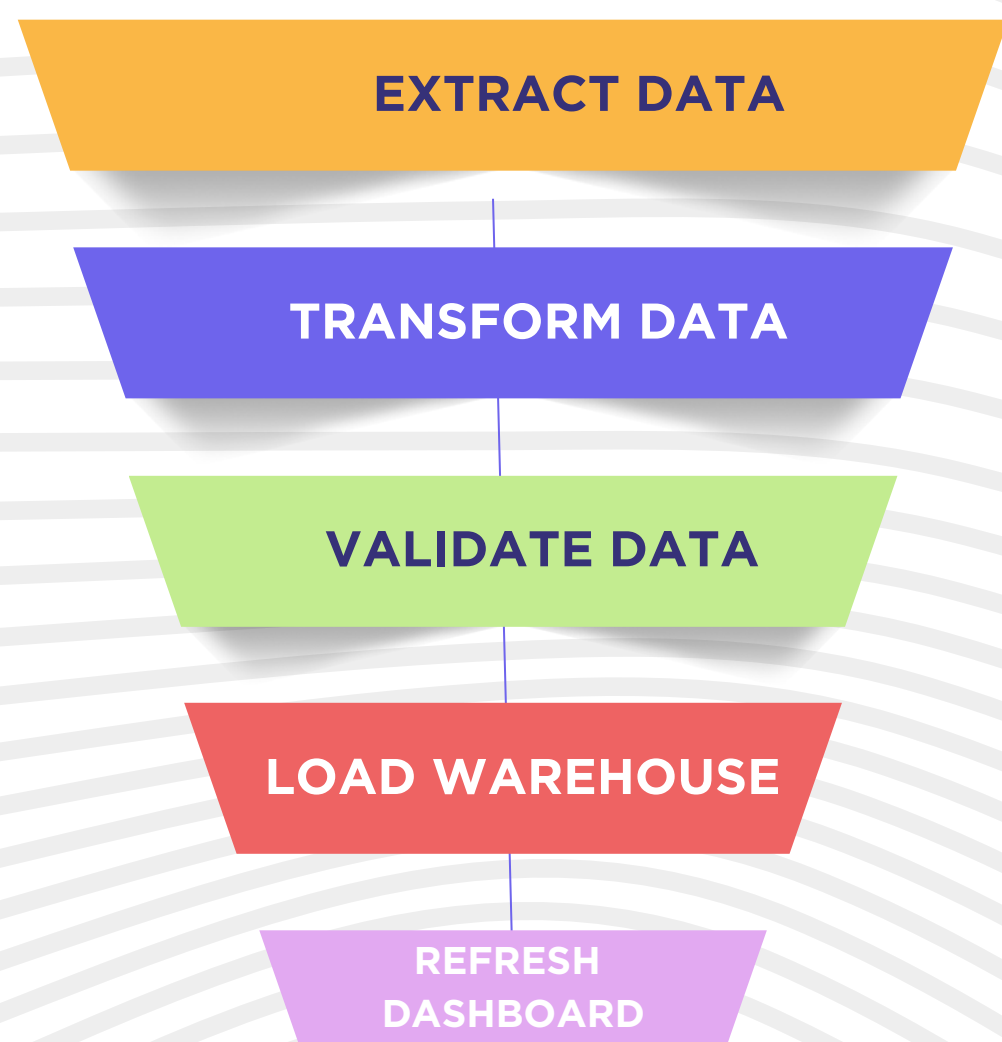


# UNDERSTANDING DAG

DAG – Directed Acyclic Graph

A DAG represents a workflow and defines task .

DAGs are the heart of Airflow.



## **DIRECTED**

Operations follow  
a specific direction



## **ACYCLIC**

No circular  
dependencies  
exist



## **GRAPH**

Represents workflow  
as connected  
operations

# AIRFLOW OPERATORS

Operators are the building blocks of an Airflow DAG. Each operator represents a single task and defines what work should be performed within a workflow. Think of operators as individual steps in a pipeline. Each step is implemented using an Airflow Operator.

## PythonOperator

1. Executes Python functions within a workflow.

## BashOperator

2. Runs shell commands or scripts directly from Airflow.

## SQL Operators

3. Execute SQL queries against databases and data warehouses.

## EmailOperator

4. Sends automated email notifications. Keeps teams informed about pipeline status.

## SparkSubmitOperator

5. Submits Apache Spark jobs for large-scale data processing. Best For Distributed data processing workloads.

## DockerOperator

6. Runs tasks inside Docker containers. Consistency across development and production environments.

## KubernetesPodOperator

7. Launches and manages tasks as Kubernetes Pods. Dynamic resource allocation and scalability.

# SCHEDULING & EXECUTION

Airflow allows workflows to run automatically on predefined schedules.

Examples

Every Hour

Every Day

Every Week

Every Month

Benefits

- Automation
- Consistency
- Reduced manual effort
- Improved reliability

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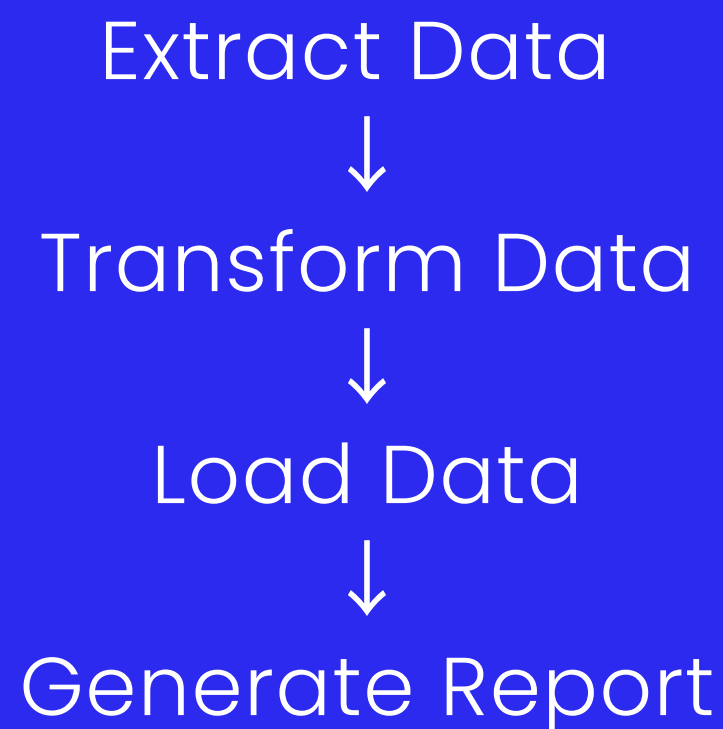


# TASK DEPENDENCIES & RETRIES

## Managing Dependencies

Airflow ensures tasks run in the correct order.

Example



If one task fails, downstream tasks won't execute.

## Retry Mechanism

Airflow can automatically retry failed tasks.

### Benefits

- Handles temporary failures
- Improves pipeline reliability
- Reduces manual intervention

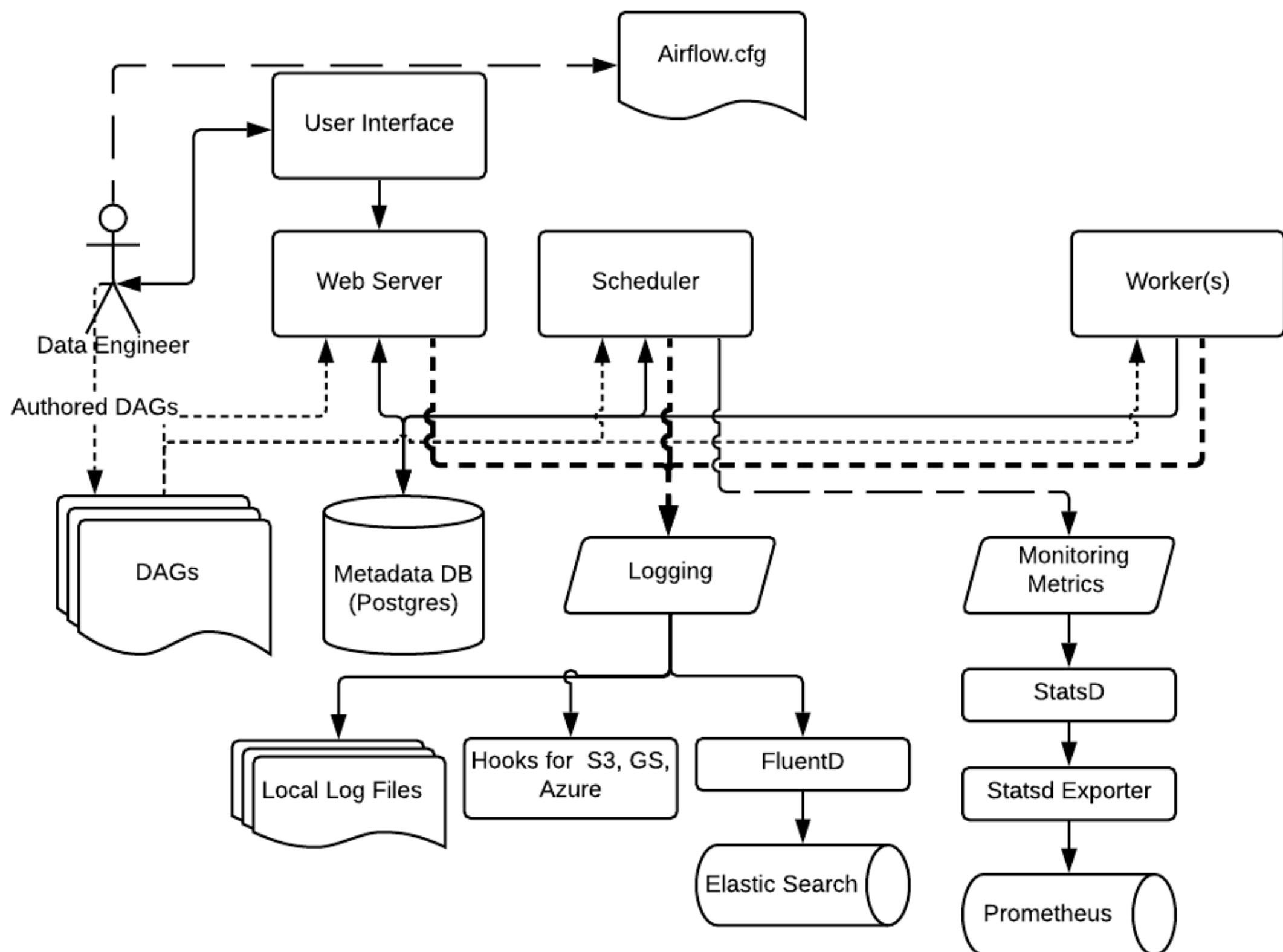
Goal:

Build fault-tolerant pipelines.

# MONITORING & OBSERVABILITY

## Airflow UI Features

- DAG Visualization – View workflow structure visually.
- Task Status Tracking – Monitor running, successful, and failed tasks.
- Execution Logs – Debug failures quickly.
- Historical Runs – Analyze previous executions.
- Alerts & Notifications – Get notified when workflows fail.



# WHY EVERY DATA ENGINEER SHOULD LEARN APACHE AIRFLOW

## 8. Open Source & Cloud Agnostic

Airflow works across AWS, Azure, GCP, and on-premise environments.

The skills you learn are transferable across different platforms.

## 7. Powerful Monitoring & Visibility

Its web interface provides real-time workflow tracking and execution logs.

Engineers can quickly identify and troubleshoot pipeline issues.

## 6. Essential for Production Pipelines

Building pipelines is only half the job; managing them is equally important.

Airflow provides scheduling, monitoring, and operational control.

## 5. Integrates with Modern Data Tools

Airflow works seamlessly with Spark, Kafka, dbt, Snowflake, and Databricks. It acts as the orchestration layer across the entire data stack.

## 1. Industry Standard

Apache Airflow is one of the most widely used workflow orchestration tools in the data industry. Many companies rely on it to manage and automate their data pipelines.

## 2. Automates Data Workflows

Airflow eliminates manual execution by automatically running tasks on schedules. This improves efficiency and reduces operational effort.

## 3. Manages Dependencies

It ensures tasks run in the correct order based on defined dependencies. This prevents failures caused by incorrect execution sequences.

## 4. Handles Failures Automatically

Airflow supports retries, alerts, and failure recovery mechanisms. This helps maintain reliable and fault-tolerant pipelines.

The background of the image consists of numerous thin, light gray lines that flow in a wavy, undulating pattern across the entire frame. These lines create a sense of movement and depth, resembling a stylized topographical map or a liquid surface. The lines are more densely packed in some areas and more spread out in others, contributing to the overall organic feel of the design.

**THANK YOU**